

# Unit A3: Key Concepts Cheat Sheet

## Database Models & Terms

- **Relational Database:** Organized collection of data in tables with relationships.
- **DBMS (Database Management System):** Software that interacts with end-users, applications, and the database itself to capture and analyze data (e.g., MySQL, PostgreSQL, Access).
- **Redundancy:** Unnecessary repetition of data. leads to anomalies and wasted space.
- **Inconsistency:** When the same data exists in different formats in multiple tables.
- **Data Integrity:** The accuracy and consistency of data over its lifecycle.

## Keys

- **Primary Key:** Unique identifier for a record. (Student\_ID)
- **Foreign Key:** A field that links to a Primary Key in another table. (Course\_ID in Student table)
- **Composite Key:** A Primary Key consisting of two or more attributes. (Often used in Link/Junction tables).
- **Candidate Key:** A column that meets the requirements to be a Primary Key (Unique, Non-null).

## Normalization Rules (The Mantra)

1. **1NF: Atomic Values** (No lists in cells) + **Unique Rows**.
2. **2NF: 1NF + No Partial Dependencies** (Non-key fields must depend on the *entire* primary key).
3. **3NF: 2NF + No Transitive Dependencies** (Non-key fields must depend *only* on the primary key).

## ERD Notation (Crow's Foot)

- **Entity:** Rectangle.
- **Relationship:** Line connecting entities.
- **One:** Single hash mark |.
- **Many:** Crow's foot <.
- **Mandatory:** Line |.
- **Optional:** Circle O.

## ACID Properties (Transaction Integrity)

- **A - Atomicity:** The transaction is an indivisible unit. All or nothing.
- **C - Consistency:** The transaction must bring the database from one valid state to another.

- **I - Isolation:** Transactions executing concurrently do not affect each other.
- **D - Durability:** Committed transactions are saved permanently, even if the system crashes.

## Data Warehouse Characteristics

- **Subject-Oriented:** Organized around major subjects (Customer, Product, Sales).
- **Integrated:** Data from diverse sources is unified (naming conventions, units).
- **Time-Variant:** Data is tagged with time; focuses on change over time.
- **Non-Volatile:** Data is loaded and accessed, not updated or deleted.

## Data Mining Goals

- **Association:** finding rules (market basket analysis).
- **Sequence Discovery:** finding time-based associations.
- **Classification:** assigning items to target categories.
- **Clustering:** finding natural groupings.